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Hyperbolic Value Addition and General Models of Animal Choice

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Three mathematical models of choice—the *contextual-choice model* (R. Grace, 1994), *delay-reduction theory* (N. Squires & E. Fantino, 1971), and a new model called the *hyperbolic value-added model*—were compared in their ability to predict the results from a wide variety of experiments with animal subjects. When supplied with 2 or 3 free parameters, all 3 models made fairly accurate predictions for a large set of experiments that used concurrent-chain procedures. One advantage of the hyperbolic value-added model is that it is derived from a simpler model that makes accurate predictions for many experiments using discrete-trial adjusting-delay procedures. Some results favor the hyperbolic value-added model and delay-reduction theory over the contextual-choice model, but more data are needed from choice situations for which the models make distinctly different predictions.

Over the past 40 years, the study of choice has become one of the major research topics in the field of operant conditioning. Some sense of the increasing level of interest in this topic can be obtained by examining volumes of the *Journal of the Experimental Analysis of Behavior,* one of the major journals publishing basic research in operant conditioning. In the first 2 years that the journal was published (1958–1959), about 17% of the research reports described studies on choice (experiments in which subjects could choose among two or more operant responses, each of which might lead to reinforcement). By the years 1987–1988, 39% of the journal's research reports involved experiments on choice, and the percentage increased to 47% for the years 1997–1998.

It should not come as a surprise that those who specialize in the experimental analysis of behavior are interested in choice, because it can be argued that all behavior involves choice. Herrnstein (1970) asserted that even in studying a single response in the sparse environment of an operant-conditioning chamber, it is essential to compare the reinforcement delivered for this response to the reinforcement inherent in alternative behaviors such as grooming, exploring, or resting. Of course, in the more complex world outside the operant-conditioning chamber, an animal's choices of possible behaviors are much more numerous. To predict a creature's behavior with any reasonable degree accuracy, it is probably essential to have some understanding of how it makes choices.

A variety of different procedures have been used in operant-conditioning experiments on choice, but most of these procedures can be divided into three main categories: *discrete-trial procedures, concurrent schedules,* and *concurrent-chain schedules.* In discrete-trial procedures, the subject chooses between different alternatives by making a single, brief response (e.g., a key peck for a pigeon, a lever press for a rat, or a keypress for a human). After a choice response, the subject receives the chosen alternative, and then there is usually an intertrial interval (ITI) before the next trial begins. Thus in discrete-trial procedures, the choice period is restricted to a small segment of each trial, and for the rest of the trial the subject experiences the consequences of that choice and cannot switch to the other alternative. In contrast, with concurrent schedules, each experimental session can be viewed as one continuous choice period, because at each moment the subject can respond on one alternative, switch to the other alternative, or make no response at all. Each alternative delivers reinforcers according to its own reinforcement schedule. Finally, concurrent-chain schedules include features from each of the other two procedures, because they alternate between the extended choice periods of concurrent schedules and periods during which the subject receives the consequences of its previous choices but cannot switch to the other alternative.

Although it is theoretically possible that different sets of principles describe choice behavior in these different classes of experimental procedures, a more parsimonious assumption is that the same basic principles of behavior apply in all choice situations. For that reason, I have argued that a worthwhile goal for research on choice is to develop a general model that can account for the results obtained with these different procedures (Mazur, 1991a). Taking on this challenge, Grace (1994, 1996) demonstrated that his *contextual-choice model* (CCM) can account for a wide range of results from experiments on choice that used discrete-trial procedures, concurrent schedules, and concurrent-chain schedules. Because of its generality, CCM is an important development in the analysis of choice. However, this article shows that two other models, Fantino's (1969) *delay-reduction theory* (DRT) and a new model, the *hyperbolic valued-added model* (HVA), can also account for a wide range of results from discrete-trial, concurrent, and concurrent-chain procedures. This article also examines some empirical results that can be used to evaluate and compare the predictions of the three models.

The next section provides an overview of experiments on choice that my colleagues and I have conducted using a discrete-trial adjusting-delay procedure. This research has demonstrated that a

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fairly simple equation called the *hyperbolic-decay model* can provide a good account of subjects' choices in these experiments. Subsequent sections show how the hyperbolic-decay equation can be generalized so that it applies to concurrent and concurrent-chain schedules, and the predictions of this model are compared with those of CCM and DRT.

Discrete-Trial Choice Experiments

Delay-Amount Trade-Offs

In its simplest form, the hyperbolic-decay model states that as a reinforcer's delay increases, its value declines according to the following equation:

$$V = \frac{A}{(1 + KD)}. \qquad (1)$$

V is the value of a reinforcer that is delivered after some delay, D; A is the value of the reinforcer if it were delivered immediately; and K is a parameter that determines how quickly value decreases with increasing delay. *Value* refers to the strength or effectiveness of a reinforcer—its ability to support choice responses, or, more generally, any type of instrumental responding. In this model, value is similar to the economic concept of subjective utility (e.g., Samuelson & Nordhaus, 1985). Economists typically assume that in a choice situation, a consumer will select the commodity with the highest subjective utility, and the hyperbolic-decay model assumes that an individual will choose whichever reinforcer has the highest value at the moment a choice is made. Economists also assume that the subjective utility of a commodity decreases with delay, and the hyperbolic-decay model states that reinforcer value decreases with delay. Classic economic theory has typically used an exponential function to describe the temporal discounting of utility. However, based on empirical results from studies with human participants, many recent economic analyses have adopted the hyperbolic function described by Equation 1 (e.g., Laibson, 1997; Lowenstein & Prelec, 1992; Overton & MacFayden, 1998).

In previous articles in which delay and amount of reinforcement were the independent variables, I have stated that the parameter A reflects the amount of reinforcement. However, as other writers have noted (e.g., Raineri & Rachlin, 1993), it is more appropriate to define A as the undiscounted value of the reinforcer (its value if delivered immediately). Several factors besides a reinforcer's amount (such as the type of reinforcer and its quality and the individual's preferences and level of motivation) will affect the reinforcer's undiscounted value.

One experiment that provided support for this model used an adjusting-delay procedure with pigeons to compare several different possible delay-of-reinforcement functions (Mazur, 1987). Figure 1 illustrates the procedure used in this experiment. A response on the center key started each trial, and then a pigeon chose either a standard alternative (by pecking the red key) or an adjusting alternative (by pecking the green key). For the condition illustrated in Figure 1, the standard alternative delivered 2 s of access to grain after a 10-s delay, and the adjusting alternative delivered 6 s of access to grain after an adjusting delay. Each reinforcer was followed by a 60-s ITI. If a pigeon chose the adjusting alternative on two consecutive trials, the adjusting delay was increased by 1 s. Conversely, if a pigeon chose the standard alternative on two consecutive trials, the adjusting delay was decreased by 1 s. The purpose of these adjustments was to estimate an indifference point—an adjusting delay at which the pigeon chose the two alternatives about equally often. Each condition of 64 trials continued for a minimum of 12 sessions and until several stability criteria were met, and the mean adjusting delay from the last 6 sessions was treated as the indifference point. For the condition illustrated in Figure 1, the mean adjusting delay for the 4 pigeons was about 30 s, suggesting that 2 s of food delivered after 10 s was about equally preferred to 6 s of food delivered after 30 s.

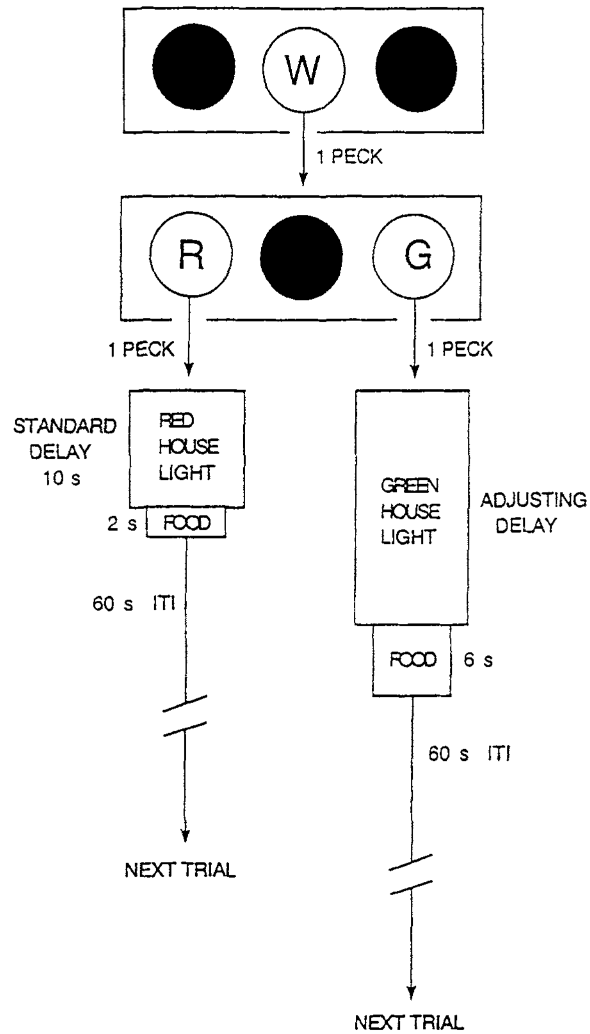

Figure 1. The two sequences of events that could occur on a choice trial in the experiment of Mazur (1987), depending on whether the red (R) or green (G) key was chosen. ITI = intertrial interval; W = white key.

Across a series of conditions, the standard delay was varied between 0 and 20 s, and for each standard delay an indifference point was obtained. Figure 2 shows the indifference points for each pigeon. As the standard delay was increased, the mean adjusting delay at the indifference point also increased, and the results were well described by linear functions with slopes greater than 1 and y-intercepts greater than 0. Mazur (1987) showed that these functions are predicted by Equation 1, but, as long as only one value of

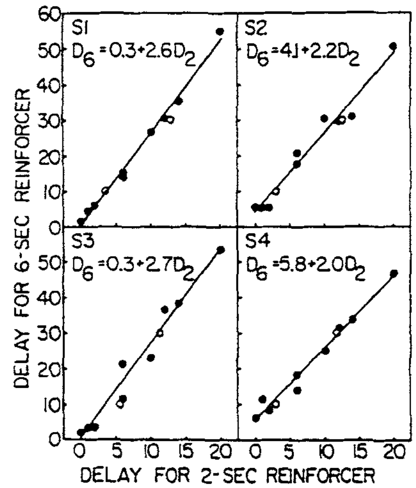

Figure 2. Indifference points (mean delays for the 6-s reinforcer) are shown for each pigeon in the experiment of Mazur (1987). Regression lines were fitted to the data from each pigeon. Note that 0 s on the x-axis is displaced to the right so that the data points can be seen more clearly. S = subject; D = delay. From "An Adjusting Procedure for Studying Delayed Reinforcement," by J. E. Mazur, 1987. In M. L. Commoms, J. E. Mazur, J. A. Nevin, & H. Rachlin (Eds.), *Quantitative Analyses of Behavior: Vol. 5. The Effect of Delay and of Intervening Events on Reinforcement Value* (pp. 55–73). Hillsdale, NJ: Erlbaum. Copyright 1987 by Lawrence Erlbaum Associates. Reprinted with permission.

K is used for each animal, the results are not consistent with several other possible delay-of-reinforcement functions, including an exponential equation, $V = A \exp(-KD)$, and a simple reciprocal equation, $V = A/KD$. These results suggest that the hyperbolic equation provides a good description of the relation between a reinforcer's delay and its value.

Other studies with pigeons (Mazur, 1986a; Rodriguez & Logue, 1988), rats (Mazur, Stellar, & Waraczynski, 1987; Richards, Mitchell, de Wit, & Seiden, 1997), and humans (Green, Fry, & Myerson, 1994; Myerson & Green, 1995) have also found that the relation between reinforcer delay and value can be well described by a hyperbolic equation. However, Green, Fristoe, and Myerson (1994) showed that if one makes the additional assumption that the decay parameter, K, is inversely related to the amount of reinforcement, an exponential equation can also predict the pattern of results shown in Figure 2 (linear indifference functions with slopes greater than 1 and y-intercepts greater than 0). Although this fact makes it more difficult to distinguish between the hyperbolic and exponential decay functions, two points should be made. First, on the basis of a more detailed analysis of their data from human participants, Myerson and Green concluded that their results were more consistent with a hyperbolic relation than with an exponential relation. Second, although the data from human participants do suggest an inverse relation between the amount of reinforcement and the value of K, this is not necessarily the case with nonhuman subjects. In a study with rats, Richards et al. found that estimates of K did not decrease but rather increased with increasing reinforcer amounts, a trend that is not compatible with the exponential equation.

The question as to whether other equations can account for results such as those shown in Figure 2 may not be completely resolved, but it seems clear that the hyperbolic equation provides at least a good first approximation to the results.

Variable Delays

It is well known that animals will show a preference for a variable reinforcement schedule over a fixed reinforcement schedule that delivers the same overall rate of reinforcement (e.g., Cicerone, 1976; Fantino, 1967; Sherman & Thomas, 1968). For example, if one alternative delivers a reinforcer after a delay that is 2 s half the time and 18 s half the time, this alternative will be strongly preferred over an alternative that delivers a reinforcer after a fixed delay of 10 s, even though the average delay is 10 s for both alternatives. For cases in which reinforcers are delivered after variable delays, the hyperbolic-decay model can be generalized as follows:

$$V = \sum_{i=1}^{n} p_i\left(\frac{A}{1 + KD_i}\right). \tag{2}$$

In Equation 2, V is the value of an alternative that can deliver any one of n possible delays on a given trial, and p_i is the probability that a delay of D_i seconds will occur. This equation states that the total value of an alternative with variable delays can be obtained by taking a weighted mean, with the value of each possible delay weighted by its probability of occurrence in the schedule.

An example shows how this model predicts preference for variable over fixed delays. If we assume for simplicity that both A and K equal 1, the value of a reinforcer delayed 10 s is 0.09. The values of reinforcers delayed 2 s and 18 s are 0.33 and 0.05, respectively, which average to 0.19. Because the average value of the reinforcer with variable delays is greater than that of the reinforcer with a 10-s delay, the hyperbolic-decay model predicts that the animal will choose the variable alternative. In addition, by setting V equal to 0.19 in Equation 1 and solving for D, we can see that the model predicts that the variable alternative should be equally preferred to a reinforcer delivered after a fixed delay of about 4.2 s. (This specific quantitative prediction depends on the assumption that K equals 1, but the more general prediction of a preference for the variable alternative does not.)

To test the predictions of the hyperbolic-decay model about preference for variable delays, Mazur (1984) used an adjusting-delay procedure similar to the one diagrammed in Figure 1, except that (a) the reinforcer amounts for the two alternatives were equal and (b) the standard alternatives were mixed-time (MT) schedules, which delivered reinforcers after either of two delays with equal probability. The results are shown in Figure 3, where the x-axis lists the MT schedules and the y-axis shows the indifference points, or the adjusting delays that were equally preferred to the MT schedules. For example, the right panel shows that with a mixed delay of 2 s and 18 s, the mean indifference point for the group was just over 4 s. The curves in Figure 3 show the best fitting predictions of Equation 2, which were obtained by treating K as a free parameter. The fits were good, and the model accounted for over 98% of the variance in the group means.

Probabilistic Reinforcers

In several experiments, Mazur (1989, 1991b, 1995; Mazur & Romano, 1992) showed that Equation 2 can also predict the results

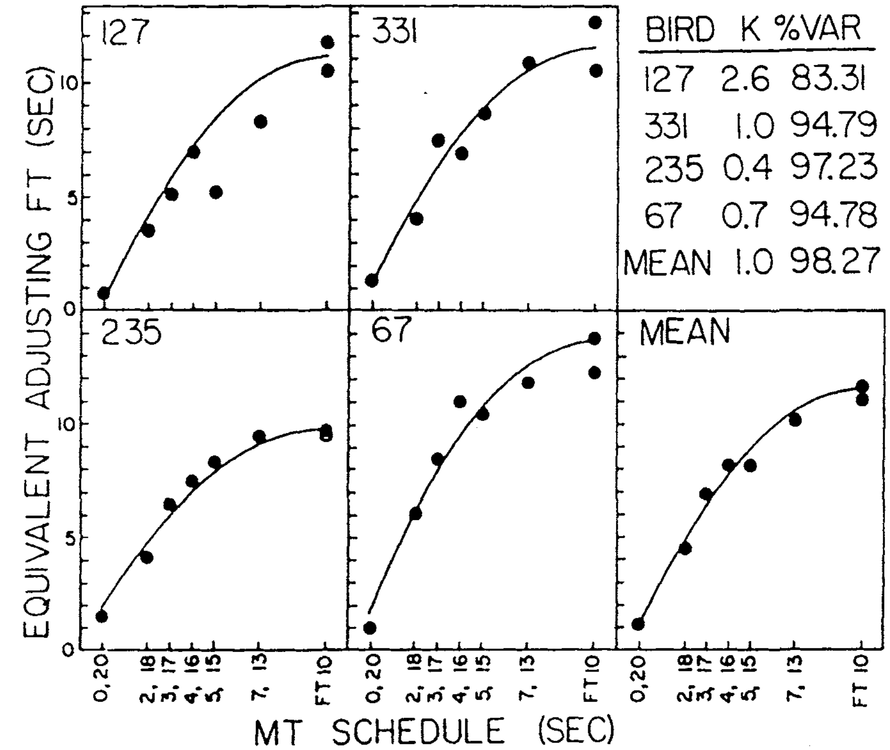

Figure 3. Indifference points from six mixed-time (MT) and two fixed-time (FT) conditions are shown for each subject in the experiment of Mazur (1984). The curves are the best fitting predictions of Equation 2, with K (a parameter that determines how quickly value decreases with increasing delay) varied as a free parameter. %Var = percentage of variance accounted for. From "Tests of an Equivalence Rule for Fixed and Variable Reinforcer Delays," by J. E. Mazur, 1983, *Journal of Experimental Psychology: Animal Behavior Processes, 10,* p. 432. Copyright 1984 by the American Psychological Association.

of discrete-trial experiments involving probabilistic reinforcement (where only a fraction of the trials end with a food delivery). Previously, Rachlin, Logue, Gibbon, and Frankel (1986) had proposed that probabilistic reinforcers are analogous to delayed reinforcers, because food is not always delivered immediately after a choice response. For example, suppose that in a variation of the adjusting-delay procedure, it takes a pigeon about 1 s to choose either the red or the green key. Each choice of the red (standard) alternative leads to a 5-s delay with red houselights, but then food is delivered on only 20% of the trials. Each choice of the green (adjusting) alternative leads to an adjusting delay, and then food is presented on every trial. All trials are separated by 30-s ITIs. Because the probability of reinforcement is .2 for the standard alternative, it will take an average of five standard trials before food is delivered. It follows that for the standard alternative, the average time between the first choice response and the delivery of food will be 150 s (which includes five 1-s response times, five 5-s delays, and four 30-s ITIs). The theory of Rachlin et al. therefore predicts that this standard alternative should be equally preferred to an adjusting alternative that delivers food on every trial after a delay of 150 s.

When Mazur (1989) conducted experiments using the procedure just described, he found that the predictions of the Rachlin et al. (1986) theory were grossly inaccurate in two ways. First, when ITI duration was varied from 30 to 90 s, this had no effect on the pigeons' indifference points, whereas according to the theory this should have made a large difference. These and other results suggest that the value of a delayed reinforcer is inversely related not to the total time between a choice response and food but to the time spent in the presence of the conditioned reinforcers that precede food delivery (in this case, the colored keylights and houselights). Mazur therefore proposed that ITI durations should not be included in the calculations. However, even when ITI durations were excluded, the actual indifference points were still shorter than predicted by the Rachlin et al. model.

As a solution to this problem, Mazur (1989) proposed that probabilistic reinforcers should be treated as reinforcers delivered after variable delays, not fixed delays. Notice that with a reinforcement probability of .2, it will take an average of five standard trials before food is delivered, but the actual number of standard trials before a food delivery will vary from one trial to many more than five trials. On the basis of this reasoning, Equation 2 was used to predict the values of probabilistic reinforcers as follows. D_i was measured as the total time spent in the presence of the conditioned reinforcers that preceded food (e.g., the colored keylights and houselights). D_i would be set to 6 s for cases in which food was delivered after the first choice of the standard alternative (to include the 1-s red keylight and the 5-s red houselights), to 12 s for cases where food was delivered after two standard choices, and so on. Values of p_i were simply the probabilities that these different runs of standard trials would occur before a food delivery. This approach has been applied to the results from several experiments on choice between probabilistic and certain reinforcers, and the predictions of Equation 2 have generally proven to be quite accurate (Mazur, 1989, 1991b, 1995).

The assumption that D_i includes only the time spent in the presence of the putative conditioned reinforcers that precede food leads to a number of predictions, some counterintuitive, that have been supported in subsequent research. For example, the model predicts that preference for a probabilistic alternative should increase if the durations of the conditioned reinforcers are decreased (thereby lowering D_i), even if the probabilities and delays for food are unchanged. Several experiments (e.g., Mazur, 1989, 1991b) have obtained this result when the conditioned reinforcers (red houselights) were simply not presented during the delays on trials without food. Conversely, the model predicts that preference for a probabilistic reinforcer should decrease if the durations of conditioned reinforcers are lengthened (again, without changing the probabilities or delays for the food). This effect has also been observed (Mazur, 1989, 1995).

These results suggest that the hyperbolic-decay model is best viewed as a method for calculating the values not of primary reinforcers themselves but of the conditioned reinforcers that precede and predict primary reinforcers. Described simply, the model states that the value of a conditioned reinforcer is inversely related to the total time spent in its presence before a primary reinforcer is delivered. Described more precisely for cases that involve variable or probabilistic reinforcers, the model states that the value of each possible delay must be calculated separately, and the total value of a choice alternative is a weighted average of these individual values.

Other Discrete-Trial Choice Phenomena

The hyperbolic-decay model has been applied to a variety of other phenomena from discrete-trial choice experiments, some of which are briefly summarized here.

Delay-ratio trade-offs. Grossbard and Mazur (1986) found results very similar to those shown in Figure 2 when fixed-ratio (FR) schedules were used in place of delays. These results showed that Equation 1 can also be applied to choices involving ratio schedules by letting D represent the number of responses in the ratio schedule or, alternatively, by letting D represent the time required to complete the ratio.

Fixed versus variable ratios. Mazur (1986b) showed that choices between fixed and variable ratios (and the preference for variable over FR schedules) can also be described by the hyperbolic-decay model. The quantitative fits were not as accurate as those for choices between fixed and variable delays, however, possibly because the actual time between a choice response and reinforcer delivery was not controlled by the procedure but depended on the subject's response rates.

Single versus multiple delayed reinforcers. Mazur (1986a) used an adjusting-delay procedure to find indifference points between a single reinforcer delivered after an adjusting delay versus one, two, or three reinforcers delivered after a series of delays. For example, in one condition, reinforcers were delivered 10, 15, and 20 s after a choice of the standard alternative. Equation 2 made accurate predictions of the indifference points, suggesting that the values of multiple reinforcers could simply be summed to obtain the total value of the standard alternative.

Variations in ITI duration. The section on probabilistic reinforcers described a study in which varying ITI duration had no effect on choice. However, Mazur, Snyderman, and Coe (1985) found that ITI duration did have an effect on choice if (a) ITI durations were different for the standard and adjusting alternatives and (b) the houselights used in the delays before food were extended into the ITIs that followed food. They showed that these effects could be predicted by the hyperbolic-decay model by assuming that reinforcers delivered on subsequent trials added to the value of the current trial, just as reinforcer values can be added when more than one reinforcer is delivered on a single trial (cf. Mazur, 1986a).

The procrastination effect. Mazur (1996, 1998) used the term *procrastination* to describe the results of experiments in which pigeons had to choose between completing a small response requirement early in a trial and completing a larger response requirement later in a trial. For example, in one condition, pigeons chose between (a) a 2-s delay, an FR 8-response requirement, a 15-s delay, then food, and (b) a 15-s delay, an adjusting response requirement, a 2-s delay, then food. On average, an indifference point was found when the adjusting response requirement was about 25 responses, which means that the pigeons were about equally likely to choose the alternative that required 8 responses near the start of a trial and an alternative that required 25 responses later in a trial (Mazur, 1996, Experiment 3). Overall, the indifference points obtained in these studies were much more variable than those obtained with positive reinforcers, but otherwise the results were similar to those from the experiments on delay-amount trade-offs. These findings suggest that delayed reinforcers and delayed response requirements may have symmetrical but opposite effects on choice and that in both cases the effects of delay may be described by a hyperbolic function.

The experiments described in this section have demonstrated that the hyperbolic-decay model can account for many different results from discrete-trial choice experiments. The next section turns to concurrent-chain procedures, which have extended choice periods in which subjects typically make many responses before a reinforcer is delivered. A method for generalizing the hyperbolic-decay model to these choice situations is presented, and its predictions are compared to those of two other models.

Concurrent-Chain Schedules

A typical concurrent-chain schedule includes a pair of schedules called *initial links,* each of which leads to its own *terminal link.* Each terminal link is another reinforcement schedule that leads to food or some other primary reinforcer. An example of a concurrent-chain schedule, one that might be used for pigeons in an operant-conditioning chamber with two response keys, is illustrated in Figure 4. In the initial links, both response keys are transilluminated with white light, and two identical variable-interval (VI) 60-s schedules are in effect. When one of the VI schedules has timed out, a response on the appropriate key leads to the terminal link for that alternative. If the left terminal link is entered, the right key becomes dark and inoperative, the left key turns green, and food is delivered on a fixed-interval (FI) 30-s schedule. If the right terminal link is entered, the left key becomes

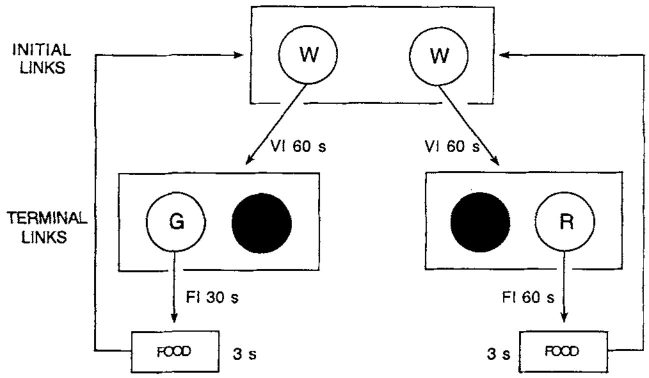

Figure 4. A typical concurrent-chain procedure, with variable-interval (VI) 60-s schedules as initial links and two different fixed-interval (FI) schedules as terminal links. G = green key; R = red key; W = white key.

dark and inoperative, the right key turns red, and food is delivered on an FI 60-s schedule. After food is delivered in either terminal link, the white keys are reinstated, and the initial links are again in effect.

In a concurrent-chain schedule, the initial links can be considered the choice period, and the ratio (or proportion) of response rates on the two keys during the initial links is used to measure the subject's preference. Numerous experiments have been conducted using concurrent-chain schedules, and a variety of behavioral phenomena have been observed. Two such phenomena are the *initial-link effect* and the *terminal-link effect.* The initial-link effect is the finding that as the durations of the VI schedules in the initial links are increased, preference for the shorter of the two terminal links (e.g., the FI 30-s schedule in Figure 4) becomes less extreme (Fantino, 1969). Conversely, the terminal-link effect is the finding that preference becomes more extreme if the durations of the two terminal links are both multiplied by the same factor (MacEwen, 1972; Williams & Fantino, 1978).

Any viable model of concurrent-chain performance must account for these effects as well as a wide range of other phenomena. A number of different mathematical models of concurrent-chain performance have been proposed. Davison (1987) examined three such models: DRT (Squires & Fantino, 1971), incentive theory (Killeen, 1982), and a variation of Herrnstein's (1961) matching law proposed by Davison and Temple (1973). Davison applied these three models to the data from 10 published experiments on concurrent-chain schedules. No free parameters were used, and the predictions of each model were compared with the group means from each experiment. Davison found that each of the three models accounted for slightly more than 50% of the variance in these 10 data sets. These percentages are not very good, especially because the models were fitted to the group means, not to the data from individual subjects. However, it was not clear at the time whether any other model could provide a better account of the data from these experiments.

CCM

Grace (1994) presented a new theory of concurrent-chain performance, CCM, and he applied this model to the same 10 data sets used by Davison (1987). CCM accounted for an average of about 91% of the variance in these data sets. This percentage is much higher than for the other three models, but in applying CCM to the various data sets, Grace included between two and four free parameters that were varied to obtain the best possible fits. It is therefore not surprising that CCM could account for a higher percentage of the variance. Nevertheless, Grace's analysis was an important advance, because it showed that at least one model could provide a good account of the results from a varied set of experiments on concurrent-chain schedules. In addition, Grace went on to apply his model to a larger group of concurrent-chain experiments, including a total of 92 data sets from individual subjects in 19 published experiments. Even when applied to the data from individual subjects, CCM still accounted for an average of about 90% of the variance.

CCM can be expressed as follows:

$$\frac{B_1}{B_2} = b\left(\frac{r_{i1}}{r_{i2}}\right)^{a_i}\left(\frac{r_{t1}}{r_{t2}}\right)^{a_t(T_t/T_i)^k}. \tag{3}$$

In this equation, 1 and 2 represent the two alternatives. B_1 and B_2 are response rates during the initial links. The quantities r_{i1} and r_{i2} are the rates of reinforcement in the two initial links (i.e., the rates at which each of the two terminal links are entered). Similarly, r_{t1} and r_{t2} are the rates of reinforcement in the two terminal links (the rates at which the terminal links deliver food). The assumption that the choice ratio, B_1/B_2, will be a function of the reinforcement ratios is an extension of Herrnstein's (1961, 1970) matching law (see Equation 7 in this article), which states that an animal will distribute its responses among different choice alternatives so as to match the distribution of reinforcers delivered by those alternatives. However, CCM is more complex than the basic matching law, because it assumes that both the initial-link and terminal-link reinforcement schedules will affect an animal's choice responses.

In CCM, the effects of the initial-link and terminal-link reinforcement rates are modulated by several free parameters. The parameter b is included to account for any possible bias a subject might have for one alternative or the other, which could result from a position preference, a color preference, or the like. The exponent a_i reflects a subject's sensitivity to differences in the initial-link schedules, and a_t reflects sensitivity to differences in the terminal-link schedules. Factors that change the discriminability of the two schedules in one link should affect the size of the corresponding sensitivity parameter. For instance, a_t should be smaller if the two terminal links are signaled by the same stimulus (e.g., blackout) than if they are signaled by distinctly different stimuli (e.g., red vs. green lights). Similarly, factors that affect the discriminability of the two initial links should alter the value of a_i.

T_t is the average terminal-link duration, T_i is the average initial-link duration, and k is a scaling parameter (which Grace, 1994, held constant at 1 for all but 10 of the 92 data sets that he analyzed). The most distinctive feature of CCM is its assumption, as expressed in the exponent T_t/T_i, that the relative importance of the two links depends on their relative durations. If the terminal-link schedules are short compared with the initial-link schedules, T_t/T_i will be small, so any difference between the two terminal-link schedules will have relatively little effect on choice. Conversely, if the terminal-link schedules are long compared to the initial-link schedules, T_t/T_i will be large, and any differences between the two terminal-link schedules will have a greater effect on choice. It is this feature of the CCM that allows the model to predict the initial-link and terminal-link effects. Stated in a different way, the CCM assumes that the animal compares the durations of the initial and terminal links, and, when making its choice responses, it gives greater weight to whichever link has the greater average duration.

Grace's (1994) analysis clearly showed that the CCM can account for the results of many experiments on concurrent-chain schedules. An unanswered question, however, was whether other models could also make accurate predictions for these experiments if they had the same number of free parameters. To address this question, I added the same number of free parameters to two other models (DRT and HVA) and applied these models to the same data sets as analyzed by Grace.

DRT

DRT is probably the most widely known theory of concurrent-chain performance. The theory was developed by Fantino (1969), and the following more general version was later presented by Squires and Fantino (1971):

$$\frac{B_1}{B_2} = \left(\frac{R_1}{R_2}\right)\left(\frac{T_{total} - T_{t1}}{T_{total} - T_{t2}}\right), \quad T_{total} > T_{t1}, \quad T_{total} > T_{t2}. \quad (4)$$

R_1 and R_2 are the overall rates of reinforcement for the two alternatives, including time spent in both the initial links and terminal links. T_{total} is the mean total time to primary reinforcement from the start of the initial links. T_{t1} and T_{t2} are the mean times to primary reinforcement from the start of the two terminal links (i.e., the average durations of the two terminal links).

Fantino (1977) described DRT as a theory about choice and conditioned reinforcement. In a concurrent-chain schedule, the stimuli associated with the terminal links can be considered conditioned reinforcers because they signal a reduction in the time to the delivery of the primary reinforcer, food. DRT states that the strength of these conditioned reinforcers is determined by the amount of delay reduction that occurs when each terminal link is entered, compared with the average time to food from the start of the initial links, T_{total}. As an example, the top panel of Figure 5 shows how DRT can be applied to the concurrent-chain schedule depicted in Figure 4, which has two VI 60-s schedules as initial links and has FI 30-s and FI 60-s terminal links. In this example, T_{total} is 75 s, because the mean time spent in the initial links is 30 s, and the mean terminal-link duration is 45 s. T_{t1} and T_{t2} are 30 s and 60 s, respectively (the durations of the two FI schedules). Therefore, there is a delay reduction of 45 s when the left terminal link is entered and a delay reduction of 15 s when the right terminal link is entered. According to DRT, the ratio of these quantities (together with the ratio R_1/R_2) determines the subject's choice ratio. For this example, DRT predicts a strong preference for the left alternative because the onset of the left terminal-link stimulus (the green keylight) signals a much greater reduction in the delay to food.

The ratio R_1/R_2 was added to DRT by Squires and Fantino (1971) to allow the model to account for the cases where the two alternatives have different rates of reinforcement as a result of unequal initial-link schedules. However, because R_1 and R_2 are the overall rates of food delivery for the two alternatives, DRT is not exclusively a theory of conditioned reinforcement: Equation 4 states that choice is determined by both the relative strengths of the conditioned reinforcers (as measured by the amount of delay reduction) and the relative rates of primary reinforcement. In this respect, DRT differs from the CCM, which includes no terms to represent the overall rates of primary reinforcement.

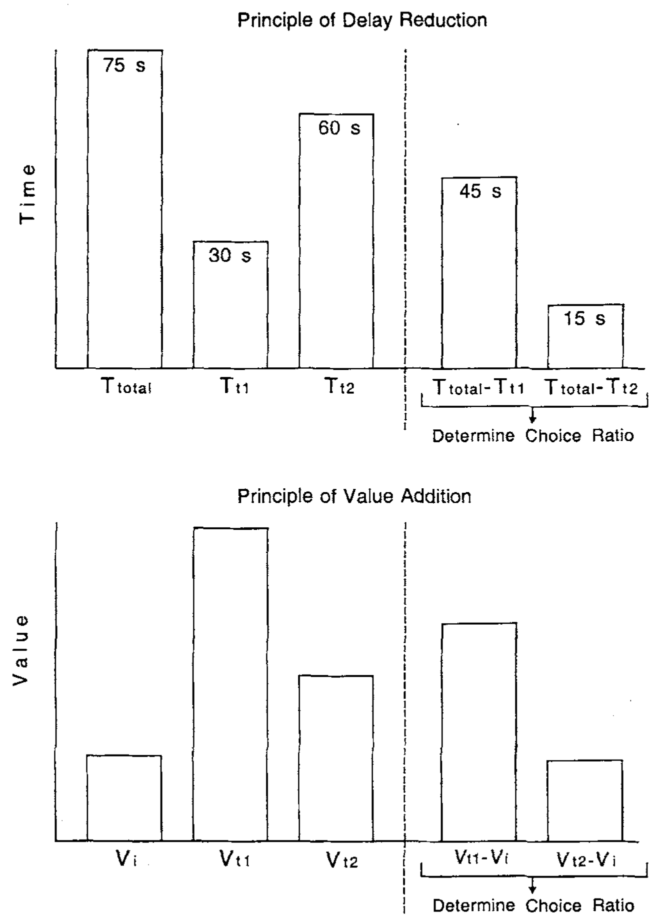

Figure 5. For the concurrent-chain procedure illustrated in Figure 4, the predictions of delay-reduction theory are compared with those of the hyperbolic value-added model. T_{total} is the mean total time to primary reinforcement from the start of the initial links; T_{t1} and T_{t2} are the mean times to primary reinforcement from the start of the two terminal links; V_i is the value of the initial links; V_{t1} is the value of the left terminal link; V_{t2} is the value of the right terminal link.

Notice that Equation 4 is applicable only if T_{total} is greater than both T_{t1} and T_{t2}. For cases in which the duration of one terminal link is greater than T_{total} (cases that are indeed possible to arrange), DRT predicts that the subject will exhibit exclusive preference for the other alternative. The reasoning behind this prediction is straightforward: DRT assumes that if there is no delay reduction when a terminal link is entered, the terminal-link stimulus will not become a conditioned reinforcer, so no choice responses will be directed toward this alternative.

The subtractive expressions in Equation 4, which describe the process of delay reduction, are what allow the theory to account for such phenomena as the initial-link and terminal-link effects. For example, if the initial-link VI schedules are lengthened, T_{total} increases, and so the ratio $(T_{total} - T_{t1})/(T_{total} - T_{t2})$ gets closer to 1. DRT therefore predicts that the subject's choice ratio will approach indifference. Conversely, if the terminal-link durations are long, T_{t1} and T_{t2} will be relatively large compared to T_{total}, and DRT predicts that preference for the shorter terminal-link schedule will be more extreme.

To make a fairer comparison between CCM and DRT, I added a comparable number of free parameters to the Squires and Fantino (1971) equation as follows:

$$\frac{B_1}{B_2} = b\left(\frac{R_1}{R_2}\right)^{a_i}\left(\frac{T_{total} - a_t T_{t1}}{T_{total} - a_t T_{t2}}\right)^k \quad T_{total} > a_t T_{t1},\ T_{total} > a_t T_{t2}. \quad (5)$$

The interpretation of the free parameters is the same as in CCM: b reflects response bias, and a_i and a_t are sensitivity parameters for the initial and terminal links, respectively (although a_t is a multiplicative parameter rather than an exponent). A fourth parameter, k, was varied only when applying DRT to the same 10 data sets for which Grace (1994) varied k when fitting CCM.

HVA

HVA begins with the assumption that the hyperbolic-decay model (Equations 1 and 2) describes how reinforcer value decreases with increasing delay. When applied to concurrent-chain schedules, HVA assumes (a) the value of each terminal link depends on the time from the onset of that link to primary reinforcement, (b) the value of the initial links depends on the (variable) time from the onset of the initial links to primary reinforcement, and (c) choice proportions are based on the amount of *value added* when a terminal link is entered (i.e., on the amount of increase in value when the terminal link is entered).

The principle of value addition is similar in many ways to the principle of delay reduction, as can be shown by again using concurrent-chain schedule in Figure 4. The predictions of HVA are illustrated in the bottom panel of Figure 5. V_i, the value of the initial links, is relatively low, because it is a relatively long time to food from the start of the initial links. V_{t2}, the value of the right terminal link, is higher, because food is 60 s away when this link begins. V_{t1}, the value of the left terminal link, is highest, because food is only 30 s away when this link begins. According to the principle of value addition, the value of the initial links is subtracted from the value of each terminal link to obtain the amount of value added when each terminal link is entered, and it is these two quantities, $(V_{t1} - V_i)$ and $(V_{t2} - V_i)$, that determine the subject's choice ratio.

In some of its basic assumptions, HVA has similarities to comparator theories of classical conditioning (e.g., Gibbon & Balsam, 1981; Miller & Schachtman, 1985). In essence, comparator theories assume that a conditioned stimulus (CS) will only elicit a conditioned response (CR) if the excitatory strength of the CS is greater than the strength of the contextual stimuli (i.e., the stimuli that are always present in the conditioning chamber). For instance, if the probability of an unconditioned stimulus is .2 both in the presence of the CS and in its absence, comparator theories assume that the CS and the contextual stimuli will have equal excitatory strengths, and they correctly predict that the CS will elicit no CR. However, if the excitatory strength of the contextual stimuli is later reduced through an extinction procedure, presenting the CS will elicit a CR, because its excitatory strength is now greater than that of the contextual stimuli (Matzel, Brown, & Miller, 1987). If the initial links of a concurrent-chain schedule are viewed as the context in which terminal-link stimuli occasionally occur, the parallel to HVA should be clear: Comparator theories assume that a CS will elicit CRs only to the extent that it has greater excitatory strength than the contextual stimuli; HVA assumes that a terminal link will attract choice responses only to the extent that it has greater value than the initial links.

HVA, with free parameters to match those of CCM, can be expressed as follows:

$$\frac{B_1}{B_2} = b\left(\frac{r_{i1}}{r_{i2}}\right)^{a_i}\left(\frac{V_{t1} - a_t V_i}{V_{t2} - a_t V_i}\right) \quad V_{t1} > a_i V_i, \quad V_{t2} > a_t V_i. \quad (6)$$

The left portions of the equation are identical to CCM, including the ratio of initial link reinforcement rates, r_{i1} and r_{i2}; a bias parameter, b; and a parameter to reflect sensitivity to the initial-link schedules, a_i. The right parenthetical expression represents the process of value addition, and it includes a multiplicative sensitivity parameter, a_t, in place of the exponent for terminal-link sensitivity in CCM. All values (V_{t1}, V_{t2}, and V_i) are calculated using the hyperbolic-decay equation (Equation 2), and the decay parameter in this equation, K, can be varied whenever k is used as a free parameter in the other two models. Similar to the restrictions on DRT, Equation 6 is only applicable when V_{t1} and V_{t2} are both greater than $a_t V_i$. HVA assumes that if the value of either terminal link is less than $a_t V_i$, the subject will show exclusive preference for the other alternative.

The process of value addition described in Equation 6 allows HVA to account for phenomena such as the initial-link and terminal-link effects in a way that is similar to DRT. As initial-link durations increase, V_i decreases, and as a result the ratio in the right parenthetical expression becomes less extreme (so the model predicts a shift toward indifference). If both terminal-link durations are multiplied by the same factor, differences between initial-link and terminal-link values decrease, and the subtraction of values in Equation 6 causes the ratio to become more extreme (so the model predicts a shift toward exclusive preference of the terminal link with higher value).

A Quantitative Comparison of the Three Models

A curve-fitting program was used to find the least squares fits of DRT and HVA (as described in Equations 5 and 6, respectively) for the same 92 data sets from the 19 experiments analyzed by Grace (1994). The program used an iterative procedure to vary each parameter over a wide range of values, progressively reducing the increments of the parameter values tested until each estimate was accurate to two decimal places. For each data set, the same number of free parameters was varied as in Grace's analyses. The curve-fitting program was also used for CCM with a sample of the data sets, to make sure that the program obtained the same results that Grace did. In all the cases examined, the program obtained parameter estimates for CCM that were identical to those obtained by Grace.[1]

Fitting DRT to these data sets involved a straightforward application of Equation 5. For each experiment, the durations of the initial and terminal links reported in the article were used to calculate R_1, R_2, T_{total}, T_{t1}, and T_{t2}, and then the curve-fitting program calculated the best fitting values of the two, three, or four parameters varied by Grace (1994) in his analyses.

Applying HVA to these data sets was not as simple, for two reasons. First, it was not obvious what value to use for the decay parameter K in Equation 2. To decide this matter, HVA was first applied to the 10 data sets for which Grace (1994) varied k. Based on the range of values obtained for K with these data sets, K was set at 0.2 for the remaining 82 data sets. Second, whereas calculating the values of the terminal links from Equation 2 was not difficult, V_i could only be calculated once the distribution of initial-link durations was estimated (because the actual distributions of these durations were not reported in any of the articles). For articles that reported the distribution of the intervals that constituted the initial-link VI schedules, this information was used to estimate the distribution of initial-link durations. For articles that did not report this information, the method described by Fleshler and Hoffman (1962) was used to approximate an exponential distribution of initial-link durations. Either way, this method of estimating V_i was not ideal because it relied on estimated rather than actual initial-link durations, but it was the best that could be done given the available data. The adequacy of this method of estimating V_i is discussed below.

For two of the experiments (Davison, 1976; Wardlaw & Davison, 1974), the results from individual subjects were not reported, so the three models were fitted to the group means. For the other 17 experiments, the models were applied separately to the data from individual subjects. For all 19 experiments, Table 1 presents the mean percentages of variance accounted for by each of the three models. As can be seen, CCM accounted for 90.8% of the variance, HVA for 89.6%, and DRT for 83.0%.

These results show that when given the same number of free parameters, the models account for similar percentages of the variance in these data sets. Although DRT accounted for slightly less of the variance than the other two models, the minor differences in percentages should be interpreted with caution. These differences may have little to do with the adequacy of the basic assumptions of the different models, because other factors may have affected the goodness of fit. One such factor is exactly how free parameters were added to each model. For example, in Equation 5, a_t was inserted as a multiplicative parameter, but this free parameter could have been added to DRT in many other ways (e.g., as an additive constant, as an exponent for T_{total}, as an exponent for T_{t1} and T_{t2}, etc.). Each of these variations of DRT would undoubtedly account for slightly different percentages of the variance in the data sets. The same can be said for how the free parameters were included in CCM and HVA.

Another factor that certainly had some effect on the percentages shown in Table 1 is the precise manner in which the equations were applied to the data from each experiment. As already mentioned, all calculations of V_i in HVA were based on approximations because the exact distributions of initial-link times were not available. Furthermore, because of the timing and pattern of a subject's responses, the mean durations of the initial and terminal links in concurrent-chain procedures are not exactly the same as the scheduled durations, yet for all three models, the scheduled durations were used in the calculations for most of the experiments because the actual durations were not reported.

Table 1
Percentages of Variance Accounted for by Three Models of Concurrent-Chain Performance

Experiment	Free parameters	CCM	HVA	DRT
Alsop & Davison, 1988	3	90.6	90.7	91.0
Chung & Herrnstein, 1967	2 or 3	91.0	90.5	83.7
Davison, 1976	3	98.2	95.6	92.5
Davison, 1983	4	81.5	82.6	61.1
Davison, 1988	2 or 3	91.6	88.7	84.0
Davison & Temple, 1973	2	92.0	90.4	78.9
Duncan & Fantino, 1970	2	98.1	95.2	86.2
Dunn & Fantino, 1982	2	99.6	94.6	99.3
Fantino, 1969	2	99.1	97.2	98.1
Fantino & Davison, 1983	3	92.1	91.4	84.3
Fantino & Royalty, 1987	2	81.0	78.3	76.4
Gentry & Marr, 1980	2 or 3	78.1	80.4	62.7
Killeen, 1970	2	98.5	98.1	97.9
MacEwen, 1972	2	92.0	93.1	88.8
Omino & Ito, 1993	2	95.4	93.4	80.2
Preston & Fantino, 1991	3	75.2	72.5	68.5
Squires & Fantino, 1971	2	85.9	85.9	68.6
Wardlaw & Davison, 1974	2	93.9	93.0	91.8
Williams & Fantino, 1978	2	90.9	90.8	83.3
M		90.8	89.6	83.0

Note. CCM = contextual-choice model; HVA = hyperbolic value-added model; DRT = delay-reduction theory.

Besides the percentages of variance that the three models accounted for, the distribution of their free parameters can also be informative. For the 92 data sets, Figure 6 presents frequency distributions of the best fitting values of the bias parameter, b, for each model. Because of individual differences in side or key color preferences, bias estimates should vary around an average value of about 1. Figure 6 shows that the distribution of b across subjects was fairly similar for all three models, and the median estimates of b were 1.13, 1.12, and 1.18 for CCM, HVA, and DRT, respectively.

Distributions of the initial-link sensitivity parameter, a_i, were also similar for the three models, which is not surprising because this parameter was used in a similar way in the different models. However, distributions of the terminal-link sensitivity parameter, a_t, were quite different for the three models. For each of the models, the default value for a_t is 1, because when a_t equals 1 it has no effect on the calculations. Figure 7 shows the distributions of a_t from the 87 data sets that included variations in terminal-link schedules. The median estimates of a_t were 1.15, 0.92, and 0.80 for CCM, HVA, and DRT, respectively.

For HVA, 82% of the estimates of a_t were between 0.5 and 1.5. The fact that most of the estimates were close to 1 suggests that the

[1] Randy Grace generously supplied me with diskette versions of the 92 data sets that included both data from the original articles and his parameter estimates for CCM, so we can be certain that the data used in my analyses of DRT and HVA were identical to those used in Grace's (1994) analyses.

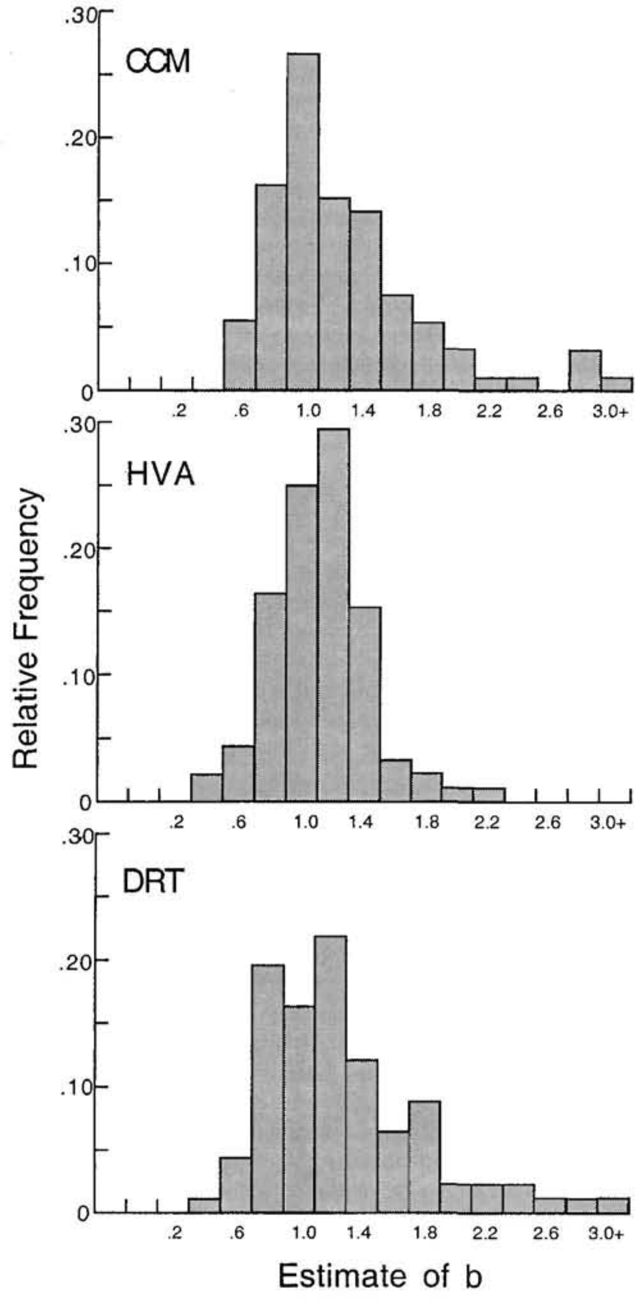

Figure 6. Frequency distributions of the bias parameter, *b*, are shown for the contextual-choice model (CCM), the hyperbolic value-added model (HVA), and delay-reduction theory (DRT) for 92 data sets from the 19 experiments listed in Table 1.

method used to estimate V_i (though based on some questionable assumptions as already discussed) was not grossly inappropriate. If there were serious problems with the calculation of V_i, estimates of a_t might have either deviated systematically from 1 or varied greatly across subjects, because the possible values of a_t were unconstrained.

Figure 7 shows that estimates of a_t were slightly more variable for DRT and much more variable for CCM. For these two models, respectively, 74% and 40% of the estimates were between 0.5 and 1.5. The wide range of estimates obtained for CCM is a possible problem. It raises questions about the suitability of CCM's method for dealing with the effects of terminal-link durations, especially because variations in CCM's estimates of a_t were not always related in a predictable way to procedural differences among the experiments. Grace (1994) noted that the average value

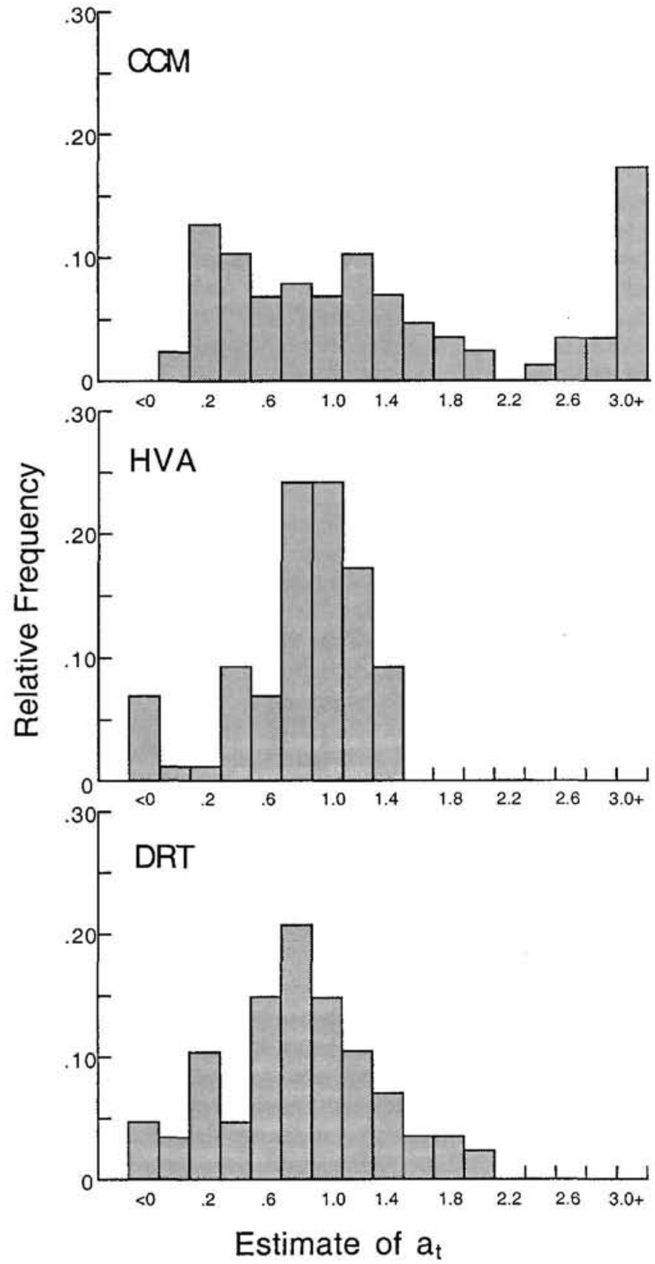

Figure 7. Frequency distributions of sensitivity to differences in the initial-link schedules, a_t, are shown for the contextual-choice model (CCM), the hyperbolic value-added model (HVA), and delay-reduction theory (DRT) for 87 data sets from 18 of the 19 experiments listed in Table 1. (For the experiment of Squires & Fantino, 1971, a_t was held constant at 1, and these data sets are not included in the frequency distributions.)

of a_t was higher for the studies that used fixed rather than variable terminal links, but there is nothing in CCM that predicts this difference. Another procedural difference among the studies was how the two terminal links were cued. Fourteen of the experiments listed in Table 1 used terminal-link stimuli similar to those illustrated in Figure 4—the two terminal links were signaled by different key colors on separate keys. Three of the other experiments used identical stimuli (blackouts) for the two terminal links, and 1 used two white keys. If a_t reflects sensitivity to terminal-link differences, we might expect to see higher estimates of a_t for the 14 studies that used different key colors than for the other 4 studies. CCM's mean estimate of a_t was somewhat higher for these 14 studies than for the other 4 (2.26 vs. 1.48). However, there was substantial overlap in the distributions of estimates for the two types of terminal-link stimuli, with mean estimates of a_t for the 14 studies ranging from 0.19 to 5.40 despite the similar arrangements of terminal-link stimuli. For comparison, DRT's mean estimates of a_t were 0.93 for the 14 experiments with different key colors and 0.85 for the other 4 studies. HVA's mean estimates of a_t were 1.00 for the 14 experiments with different key colors and 0.16 for the other 4 studies. HVA's estimates for the 4 studies that did not use different colors for terminal-link stimuli were the lowest four estimates out of all the studies listed in Table 1. Therefore, HVA's estimates of a_t showed the clearest separation between studies that used different colors for terminal-link stimuli and those that did not.

In summary, these three models, which make different assumptions about the mechanisms of choice behavior, can each provide respectable accounts of the results from a large set of experiments on concurrent-chain schedules. CCM's estimates of a_t are a possible source of concern, however, because they varied widely across experiments in ways that were not always related to the procedures used in the different studies.

Application to Other Phenomena

For any model to be considered a general model of choice, it should be able to account for findings from a variety of different choice procedures. This section will review several such findings and evaluate the extent to which HVA, CCM, and DRT can accommodate them.

Matching and Its Variations

Herrnstein (1961) found that when pigeons were presented with a pair of VI schedules (concurrent VI schedules), their response ratios tended to match the ratio of reinforcement rates delivered by the two schedules. Later research showed that several types of deviations from perfect matching can occur: Response ratios may be less extreme than the reinforcement ratios (*undermatching*), they may be more extreme than the reinforcement ratios (*overmatching*), or subjects may exhibit a bias for one of the two alternatives. All of these deviations from matching can be described by the *generalized matching law* (Baum, 1974), which, in the notation used in this article, can be written as follows:

$$\frac{B_1}{B_2} = b\left(\frac{r_{i1}}{r_{i2}}\right)^{a_i}. \tag{7}$$

Concurrent VI schedules can be considered a special case of concurrent-chain schedule in which the terminal-link durations are zero, so r_{i1} and r_{i2} can represent the reinforcement rates of the two VI schedules. As in the models already presented, the generalized matching law has a bias parameter and an exponent that reflects the subject's sensitivity to the reinforcement ratio. Undermatching is described by values of a_i less than 1, and overmatching by values greater than 1.

It is easy to show that HVA, CCM, and DRT all reduce to the generalized matching law when applied to concurrent VI schedules: In all three models, the expression representing terminal-link ratios becomes 1 if terminal-link durations are zero, and choice is determined by the ratio of the initial-link schedules. These models are not unique in this respect, because quite a few other theories of choice can also predict matching and its variations (e.g., Gibbon, 1977; Herrnstein & Vaughan, 1980; Killeen, 1982; Myerson & Miezin, 1980).

Nonadjusting Discrete-Trial Procedures

For simple discrete-trial choice procedures with no adjustment, all three models predict that if the duration of the choice period is short, and if the differences between the two alternatives are large enough for a subject to discriminate, choice responses will approach exclusive preference for whichever alternative has the higher value (or the shorter time to reinforcement). DRT and HVA make this prediction because as initial links are shortened, the subtractive expression for the less valued terminal link (in Equations 5 and 6, respectively) will approach zero and eventually become negative. In CCM, the exponent T_t/T_i becomes increasingly large as initial links are shortened, and so the model predicts more and more extreme preference for the shorter terminal link.

Exclusive or near-exclusive preference has sometimes been observed in discrete-trial choice procedures (e.g., Logue & Pena-Correal, 1985; Snyderman, 1983b), but often subjects show gradations of preference if one alternative is varied across conditions whereas the other is kept constant (e.g., Green, Fisher, Perlow, & Sherman, 1981; Young, 1981). Each of the three models does predict such gradations in preference if values of less than 1 are used for a_t (and possibly for a_i as well, depending on exactly how the models are applied to specific procedures). Thus, although it may be possible to arrange specific discrete-trial situations for which the models make different predictions, all three can predict the most general features of discrete-trial choice—exclusive preference in some cases and gradations of preference in others.

The Constant-Difference Effect in Concurrent Chains

Savastano and Fantino (1996) described two experiments on concurrent-chain schedules that obtained a counterintuitive result called the constant-difference effect. These experiments compared conditions in which the terminal-link schedules were of different durations but differed by the same amount. For example, in one condition the two terminal-link schedules were VI 5-s and VI 25-s schedules, and in another condition they were VI 100-s and VI 120-s schedules. Notice that in each pair, the terminal links differed by 20 s. In their experiments, Savastano and Fantino found that choice ratios remained approximately unchanged as long as the difference between the two schedules remained constant.

This result is counterintuitive because it contradicts Weber's law of difference thresholds and stimulus discrimination. Based on

Weber's law, we might expect preference to move toward indifference with the longer terminal links, because the difference between VI 100-s and VI 120-s schedules will be more difficult for subjects to discriminate. Savastano and Fantino (1996) showed that this result also conflicts with the predictions of several theories of choice (e.g., Herrnstein, 1964; Killeen & Fantino, 1990; Vaughan, 1985). They acknowledged, however, that their result might be consistent with CCM, because for a wide range of terminal-link durations, the model predicts little change in choice ratios if a constant difference between terminal-link durations is maintained.

Savastano and Fantino (1996) demonstrated that Fantino's (1969) original version of DRT (which does not include R_1 and R_2, the overall reinforcement rates) predicts a strict constant-difference effect. For example, with a mean initial-link duration of 30 s, the delay reduction is 40 s for the shorter terminal link and 20 s for the longer terminal link for any pair of terminal-link schedules that differ by 20 s. However, once R_1 and R_2 are added to DRT (as in Equation 4), the model does not predict a strict constant-difference effect but rather a gradual decline in preference toward indifference with increasing terminal-link durations. (It is worth noting that although the group means in the Savastano and Fantino experiments showed no significant changes in response ratios across conditions, 3 of 4 pigeons in both experiments did show a small decrease in response ratios between the shortest and longest pairs of terminal links.) Furthermore, if free parameters are added to DRT as in Equation 5, the rate of change in choice ratios with changing terminal-link durations depends on the parameter values. In short, the predictions of DRT regarding the constant-difference effect become less specific when the more general version described in Equation 5 is used.

Using typical values for its free parameters, HVA does not predict the constant-difference effect. Like most of the models reviewed by Savastano and Fantino (1996), HVA predicts a decline in preference with increasing terminal-link durations.

In summary, although Savastano and Fantino (1996) claimed that the constant-difference effect favors DRT over other models, this is not necessarily the case for two reasons. First, the more general version of DRT (Squires & Fantino, 1971) predicts not a strict constant-difference effect but rather a gradual decline in choice ratios. Second, the rate of decline in preference predicted by both DRT and by other models depends on the values of their free parameters. Therefore, the constant-difference effect, though certainly an intriguing result worthy of further investigation, does not necessarily constitute a critical test of these models.

Fixed Versus Variable Terminal Links

Just as in discrete-trial procedures, animals exhibit a preference for variable over fixed schedules when they are used as terminal links in concurrent-chain schedules. As DRT has been typically formulated (Equation 4), this model does not predict preference for variable over fixed schedules, either in discrete-trial or concurrent-chain procedures. This is because T_{total}, T_{t1}, and T_{t2} are all calculated as mean durations, without regard to whether the durations are fixed or variable. It might be possible to modify DRT so that it predicts preference for variable schedules. For example, geometric or harmonic means could be used in place of arithmetic means to calculate these durations (Herrnstein, 1964; Killeen, 1968). However, implementing such a strategy would immediately raise additional questions (such as whether and how to take into account initial-link variability when calculating T_{total}, and how other predictions of DRT might be changed as a result of this modification), and because Fantino has never made such a modification in DRT, it is not attempted here.

Similarly, the version of CCM described by Equation 3 does not distinguish between fixed and variable terminal links. However, Grace (1996) proposed a simple modification of CCM that allows it to accommodate this phenomenon. In place of the terminal-link response rates, r_{t1} and r_{t2}, Grace used terminal-link values, V_{t1} and V_{t2}, which were defined as follows:

$$V = \sum_{i=1}^{n} p_i\left(\frac{1}{D_i^{a_t}}\right). \tag{8}$$

This equation is similar to the hyperbolic-decay model (Equation 2) except that it does not include a constant of 1 in the denominator, and the exponent a_t is used as a free parameter instead of the multiplicative parameter K. (Because a_t is applied to each individual delay in Equation 8, it is no longer used as an exponent for the terminal-link ratios in Equation 3.)

Grace (1996) applied this version of CCM to data sets from 34 individual subjects in seven experiments that compared fixed and variable terminal links. In all cases, Grace varied two free parameters, b and a_t. To compare CCM and HVA, I applied HVA to these same 34 data sets, also varying b and a_t as free parameters. Because initial- and terminal-link values as calculated by Equation 2 already take variable durations into account, HVA could be applied to these data sets in exactly the same way that it was applied to the data sets in Table 1.

Table 2 shows the mean percentages of variance each model accounted for in each of the seven experiments. On average, CCM accounted for about 1% more of the variance, but HVA had slightly higher percentages in five of the seven studies. Thus, as in the comparisons shown in Table 1, the predictive accuracy of the two models was very similar.

Delay-Amount Trade-Offs in Concurrent Chains

An additional level of complexity can be added to concurrent-chain schedules by varying the amount of reinforcement delivered

Table 2
Percentages of Variance Accounted for by CCM and HVA in Seven Experiments on Choice Between Fixed and Variable Terminal Links

Experiment	CCM	HVA
Cicerone, 1976	93.0	96.0
Davison, 1969	94.6	94.9
Davison, 1972	88.0	88.2
Hursh & Fantino, 1973	80.6	63.4
Killeen, 1968	77.8	82.6
Navarick & Fantino, 1972	89.0	86.9
Rider, 1983	88.0	89.4
M	87.3	85.9

Note. CCM = contextual-choice model; HVA = hyperbolic value-added model.

by the two alternatives. Snyderman (1983a) conducted one such experiment. Two VI 60-s schedules were used as the initial links, and fixed-time (FT) schedules of different durations were used as terminal links, but the two alternatives delivered 2 s and 6 s of access to grain, respectively. Across conditions, Snyderman varied the durations of the FT schedules, and he examined several different ratios of large and small delays. For example, in three conditions, the delay for the larger amount of food was always 6 times longer than the delay for the smaller amount, but three different pairs of FT schedules were used (12 s vs. 2 s, 60 s vs. 10 s, and 120 s vs. 20 s).

Figure 8 shows that as terminal links were lengthened, the pigeons' choice proportions varied in different ways depending on the ratio of the terminal-link durations. In the 6:1 conditions (those with delays 6 times longer for the larger reinforcer), choice of the larger reinforcer declined precipitously with increasing delays. Conversely, in the 1:1 conditions (those with equal delays for the two reinforcers), choice of the larger reinforcer increased gradually with increasing delays. Choice proportions from the 3:2 and 3:1 conditions were between these two extremes.

CCM, HVA, and DRT can each describe the general patterns of results shown in Figure 8 if two free parameters are varied. The curves in Figure 8 are the best fitting predictions of HVA, which were obtained by varying a_t and A, the parameter that reflects reinforcer amount in Equation 2. Although the reinforcer durations differed by a factor of 3, this does not mean that the value of the larger reinforcer was 3 times as large. Several studies with pigeons have found evidence that the reinforcing value of 6 s of grain is much less than 3 times that of 2 s of grain (Grossbard & Mazur, 1986; Mazur, 1987; Rodriguez & Logue, 1988). The predictions shown in Figure 8 (obtained with a_t equal to 0.59, and A_L, the value of the larger reinforcer, equal to 1.45 times the value of the smaller reinforcer) accounted for 87.3% of the variance. Although the fits to the data were not perfect, there was considerable variability among subjects in Snyderman's (1983a) experiment, and HVA predicted the general trends in the choice proportions for all four delay ratios. The fits of the other two models were similar, with CCM and DRT accounting for 88.1% and 78.6% of the variance, respectively.

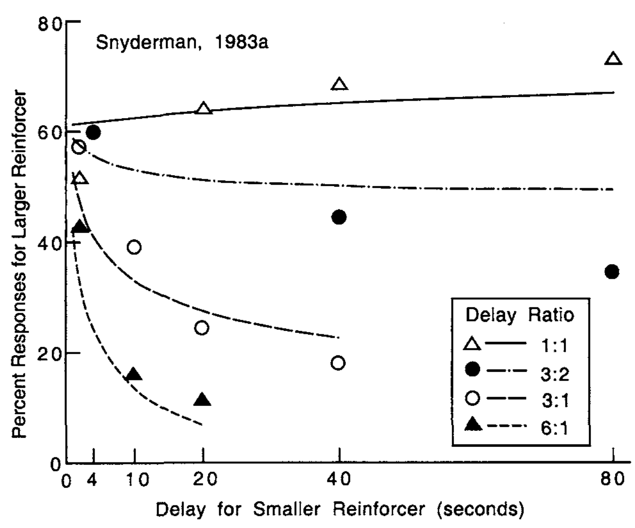

Figure 8. Response percentages for the larger reinforcer are shown for each condition of the experiment of Snyderman (1983a). The curves show the best fitting predictions of the hyperbolic value-added model. From "Delay and Amount of Reward in a Concurrent Chain," by M. Synderman, 1983, *Journal of the Experimental Analysis of Behavior, 39,* p. 446. Copyright 1983 by the Society for the Experimental Analysis of Behavior, Inc. Adapted with permission.

Two- Versus Three-Alternative Concurrent-Chain Schedules

Some of the more revealing comparisons of these models come from cases in which they make distinctly different predictions. Mazur (2000) examined several concurrent-chain procedures for which the predictions of CCM were qualitatively different from those of both DRT and HVA. Figure 9 shows one such procedure. Pigeons' choices were recorded in a three-key operant chamber, and the initial-link schedules (when operating) were VI 60-s schedules for all three keys. The initial links used a procedure known as dependent VI scheduling (Stubbs & Pliskoff, 1969), which ensured that each of the available terminal links would be entered about equally often, regardless of how a subject responded. In even-numbered sessions, only the two side-key schedules were available; the center key was dark and inoperative (see the left panel of Figure 9). The left terminal link delivered food on an FT 3-s schedule (i.e., after a 3-s delay during which no responses were required). The right terminal link delivered food on an FT 12-s schedule. In odd-numbered sessions (right panel), the schedules for the two side keys remained the same, but now the center key was lit and operative, with a VI 60-s initial-link schedule and no terminal-link delay (i.e., food was delivered immediately on terminal-link entry).

The main question of interest in this experiment was how the presence versus absence of the center-key schedule would affect the ratio of left to right responses. CCM predicts that the addition of the center-key schedule should have no effect on the left–right response ratio, because the initial-link and terminal-link schedules were unchanged for these two keys, and because the addition of the center-key schedule did not change the ratio T_t/T_i. (Without the center-key schedule, T_t/T_i = 7.5 s/30 s; with the center-key schedule operating, T_t/T_i = 5 s/20 s.)

In contrast, both HVA and DRT predict that the left–right response ratio should increase when the center-key schedule is operating. According to HVA, the value of the initial links, V_i, will now be greater, because the delays to food from the start of the initial links will be shorter when the center key is operating. Because of the greater value of the initial links, there will not be much value added when the right terminal link, with its 12-s delay to food, is entered. However, there will still be substantial value added when the left terminal link, with its 3-s delay to food, is entered. Therefore, the ratio $(V_{t1} - a_tV_i)/(V_{t2} - a_tV_i)$ will be greater when the center-key schedule is operating. For similar reasons, DRT predicts that there will not be much delay reduction when the right terminal link is entered, but there will still be substantial delay reduction when the left terminal link is entered, so the ratio $(T_{\text{total}} - a_tT_{t1})/(T_{\text{total}} - a_tT_{t2})$ will be greater.

As the presence and absence of the center-key schedule alternated in successive sessions, the left–right response ratios of all 4

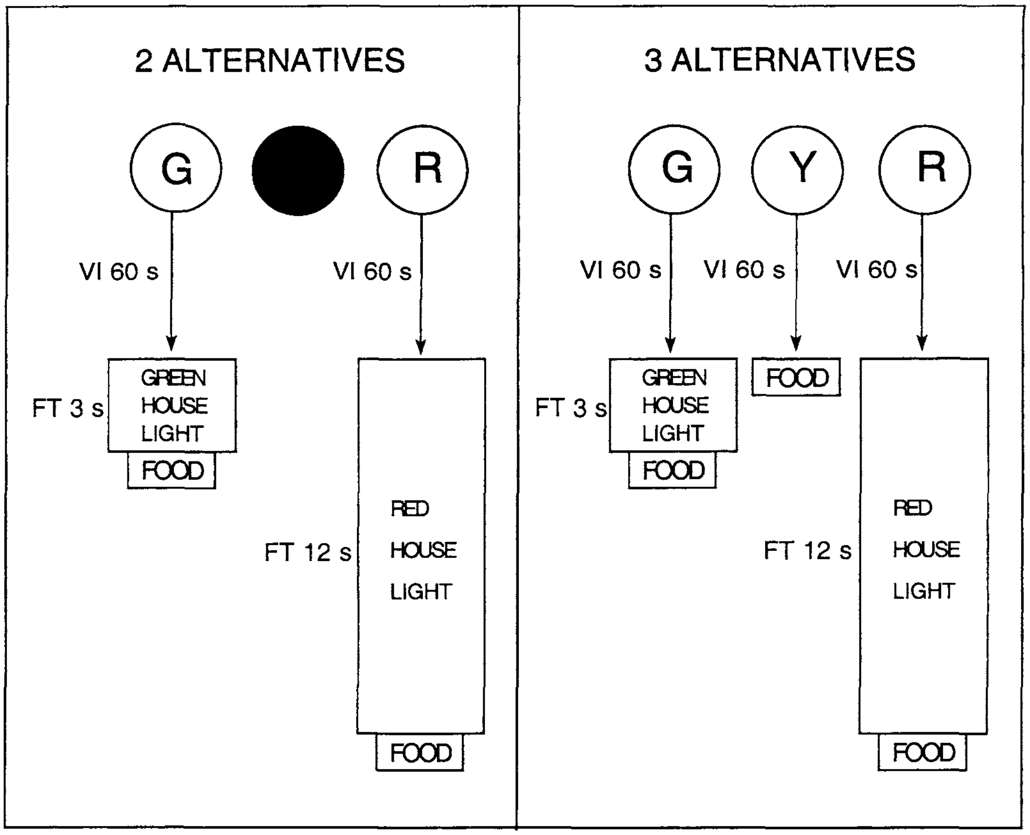

Figure 9. The procedure in one condition from the experiment of Mazur (2000). The left panel shows the procedure for sessions in which two alternatives were available, and the right panel for sessions in which three alternatives were available. G = green key; R = red key; Y = yellow key; VI = variable-interval (schedule); FT = fixed-time (schedule). From "Two- Versus Three-Alternative Concurrent-Chain Schedules: A Test of Three Models," by J. E. Mazur, 2000, *Journal of Experimental Psychology: Animal Behavior Processes, 26,* p. 289. Copyright 2000 by the American Psychological Association.

pigeons increased when the center-key schedule was present and decreased when it was not. The mean left–right response ratio in the last 10 sessions of this condition was about 14:1 when the center-key was operating and 5:1 when it was not. This difference was highly significant, and in almost every case the ratio in a session with the center key operating was higher than in the sessions that preceded and followed it. These results are consistent with HVA and DRT, but not with CCM.

Another condition in this experiment used a similar comparison of two- versus three-key concurrent chains, but with a different combination of schedules. In this case, HVA and DRT predicted a decrease in the left–right ratio when the center-key schedule was operating, whereas CCM predicted an increase. The left–right response ratio decreased for all 4 pigeons when the center-key schedule was operating. This research on two- versus three-alternative concurrent chains is informative because the predictions of CCM are opposite to those of HVA and DRT. In both conditions, the presence of the center-key schedule produced changes in preference in the direction predicted by HVA and DRT and contrary to the predictions of CCM.

Conclusions

This article has focused on three general models of animal choice, CCM, HVA, and DRT. I have called them general models because (a) each model can be applied, at least in principle, to the three main procedures used to study animal choice—discrete-trial procedures, concurrent schedules, and concurrent-chain schedules—and (b) each model can account for a variety of well-established results obtained with these choice procedures (e.g., matching and its variations, the initial-link and terminal-link effects, delay-amount trade-offs, etc.). These models represent a definite departure from earlier theories of animal choice, which could account for results from one class of procedures but not others. For example, the generalized matching law (Equation 7) is most successfully applied to results for concurrent schedules, and the hyperbolic-decay model (Equation 2) to results from discrete-trial procedures. Although attempts were made to apply these models to data obtained from other choice procedures (e.g., Chung & Herrnstein, 1967; Davison, 1988), those attempts were not very successful. In contrast, CCM, HVA, and DRT can each be successfully applied to a much wider range of research findings.

The analyses summarized in Table 1 showed that each of these three models, when supplied with two or more free parameters, can account for a large percentage of the variance in a wide variety of studies that used concurrent-chain procedures. The free parameters are necessary for any model to account for these data in quantitative detail. In a procedure as complex as a concurrent-chain schedule, factors such as bias and sensitivity to differences in the initial-link and terminal-link schedules will undoubtedly vary from individual to individual and from experiment to experiment. In most cases, there is no way to account for these differences among subjects or among experiments if a model contains no free parameters.

The three models make different assumptions about the basic determinants of choice behavior. According to CCM, the values of terminal-link schedules are independent of the initial-link schedules that precede them, but the expression of these values in the animal's choice behavior is modulated by the relative durations of the initial and terminal links (Grace, 1996). According to DRT, choice behavior is determined by the relative amount of delay reduction that occurs at the moment of transition from initial to terminal links. Finally, HVA assumes that choice behavior is determined by the amount of value added at the moment of transition from initial to terminal links.

These assumptions about the determinants of choice behavior are similar enough to allow each of the models to account for a wide range of results, and as this article has shown, the predictions of the three models are often quite similar. However, in some cases the predictions of the models are quite different, and some of the existing data favor one model over another. For instance, unlike CCM and HVA, DRT in its present form makes no distinction between fixed and variable schedules. This is a major limitation of the model, because it does not predict the preference for variable schedules that has repeatedly been found in both discrete-trial and concurrent-chain procedures. As already mentioned, there are ways that DRT could be modified to accommodate differences between fixed and variable schedules, such as using some measure other than arithmetic means to calculate time to reinforcement. However, such a modification would constitute a major change in the form of the model, because mean time to reinforcement from the start of a link is DRT's most important independent variable. It is not clear how changing this variable would affect the model's predictions about the many other phenomena to which it has been applied.

CCM is a versatile model that can make excellent predictions for a wide range of concurrent-chain schedules involving both fixed and variable terminal links, as the results in Tables 1 and 2 demonstrate. In addition, Grace (1996) showed that this model can also be applied to discrete-trial adjusting-delay procedures. However, the large degree of variability in its estimates of a_t, the terminal-link sensitivity parameter (Figure 7), may indicate a possible weakness of the model, particularly because in some cases these variations did not appear to be related to the procedures used in the different experiments. Furthermore, the results from Mazur's (2000) research on two- versus three-alternative concurrent-chain schedules were in direct contradiction to CCM's predictions, whereas they were consistent with the predictions of HVA and DRT. It would be a mistake to dismiss CCM or any model because it fails to predict a small body of data, because there could be problems with the way the data were collected and because some small change in the assumptions of the model might improve its predictions. In the long run, however, this type of choice situation (one for which different models make qualitatively different predictions) may be the best way to discriminate among the competing models.

One possible weakness of HVA is that it does not predict the constant-difference effect reported by Savastano and Fantino (1996), although the rate of decline in preference that HVA predicts can be modulated by varying the model's free parameters. Additional research may help to determine how serious a problem this phenomenon poses for HVA. On the positive side, the analyses summarized in Tables 1 and 2 show that HVA can make predictions for concurrent-chain studies involving both fixed and variable terminal links that are comparable in accuracy to those of CCM. In addition, variations in HVA's estimates of a_t appear to be more closely related to procedural differences among the experiments listed in Table 1 than are the estimates of either CCM or DRT.

Finally, an important advantage of HVA is that as initial-link durations approach zero (as they do in many discrete-trial choice procedures), the model's calculations of reinforcer values reduce to those of the hyperbolic-decay model (Equation 2). Therefore, HVA can accommodate all of the phenomena from adjusting-delay procedures described in the first part of this article just as well as the hyperbolic-decay model does. This is not necessarily the case for the other two models. As already discussed, DRT does not predict the differences in preference for fixed versus variable schedules. At present, neither DRT nor CCM makes clear predictions for the adjusting-delay experiments on probabilistic reinforcement (e.g., Mazur, 1989, 1991b, 1995; Mazur & Romano, 1992), because these models do not specify how to average over a series of trials that may or may not end with primary reinforcement. Thus one strength of HVA is the wide range of choice situations for which it makes specific predictions.

One limitation of all three of the models is that they predict only the average response ratio, B_1/B_2, over extended periods of time. They make no predictions about any local patterns of responding that may occur in the initial links of concurrent-chain situations or in other choice situations. Other theories of animal choice have addressed these matters (e.g., Dragoi & Staddon, 1999; Gibbon, Church, Fairhurst, & Kacelnik, 1988), but how the models described in this article might be extended to make predictions about local response patterns is not clear.

The extensive body of data on choice that has been obtained with discrete-trial, concurrent, and concurrent-chain procedures has clearly served to shape the forms of these general models of choice, and this is reflected in the similarities of the three models. Each of the models takes into account such factors as bias, initial-link durations, terminal-link durations, and individual differences in sensitivity to these different components of a choice situation. Yet although the models are similar in form, they are not identical, and as the experiments on two- versus three-alternative choice illustrate, it is possible to devise empirical tests for which the models make distinctly different predictions. Future research should be directed at designing and implementing additional critical tests that will help to distinguish among the predictions of these and other possible models. In future studies with concurrent-chain schedules, the collection and analysis of response sequences and distributions of initial-link durations should provide more rigorous tests of these models. Such research will further narrow down the range of possible forms a general model of choice can take and will lead to a better understanding of the mechanisms that underlie choice behavior.

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